Examiner reports

Q1.

Parts (a), (b) and (c) were completed well by the majority of candidates and most of the errors were to be found in part (d). In part (a), some candidates made errors such as omitting arrowheads or using inappropriate labels, such as 4 kg instead of $4g$. In part (d), although there were many correct answers, the candidates produced a wide range of incorrect answers. Common errors included candidates' dividing either the 30 N (from the question) or 11.76 N (the friction) by the mass; other candidates calculated the resultant force and not the acceleration.

Q2.

Parts (a), (b) and (c) of this question were done very well. In part (a), almost all candidates used an equation of motion for each particle and there were very few 'single equation' solutions. In part (c), some candidates stated one incorrect assumption, for example one about the peg rather than the string. In part (d), a few candidates took the acceleration to be equal to g rather than the value obtained in the question.

Q3.

While there were a number of good solutions to this question, there were also a very significant number of weak attempts. In part (a), many candidates could write down either a correct horizontal or vertical equation, but then were unable to derive the required answer correctly. Some could get one equation but not the other. In some cases the equations were confused, for example containing both a and g in the same equation. In part (b), there were again a number of correct solutions. The most common error here was to work on the assumption that P would also be zero. The candidates who answered correctly produced a variety of solutions, which included finding an expression for Q and starting again as if there was only one string.

Q4.

Most candidates showed that they knew the techniques involved in answering this question. Candidates generally answered part (a) correctly. Part (b)(i) was answered well, but in part (b)(ii) some candidates forgot to find the magnitude of **F**, giving instead just the vector **F**, – 40**i** + 30**j**. Most candidates answered part (c) correctly. In part (d) the majority of candidates integrated to find the position vector, but a common error was to forget the +**c** term and thus to ignore the initial position vector. Some found the + **c** incorrectly, assuming that **c** was the initial position vector.

Q5.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for ignoring the m term or ignoring the minus sign. There were many good answers to part (b), although a significant number of candidates used 'inventive' algebra. A few candidates who integrated to find the equation in v and t forgot the $+c$ term.

Q6.

Most candidates answered parts (a), (b) and (c) well, although some did get an answer of

–4.5**j** for part (b).

In part (c), the candidates who clearly stated and used a constant acceleration equation usually made a good start and completed the question, although there were a few arithmetic errors.

In part (e) many candidates did not attempt to find the magnitude of the force.

Part (f) was generally found more difficult with only a few good attempts. Several candidates did not attempt this part of the question.

Q7.

Most candidates made good attempts at this question. In part (b) (ii) there were some solutions that did not contain enough working to justify the award of marks, for example $a = \frac{4}{0.5} = 8$ with no explanation of where the 4 came from. Some also simply found a way to get 8, for example from something like 9.8 – 2 × 0.9 = 8.

For part (b) (iii) there were many correct solutions, but some candidates took the acceleration as 9.8 instead of 8. The final part proved to be more difficult for the candidates. While there were many candidates who produced good answers, there were also quite a lot who could not give a clear or relevant explanation.

Q8.

Parts (a) and (b) of this question were done very well and many candidates gained all four marks.

Part (c) proved to be more challenging. In part (c)(i), many candidates did not show how they started their working. An equation based on $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ was expected, but not always seen. Some candidates obtained the printed answer, but did not gain full marks as their starting point was not clear. Having both the acceleration and the initial velocity given in the question enabled many candidates to write down a correct expression for the velocity of the particle. Part (c)(iii) was found to be very challenging. If candidates were able to write down an expression for the speed of the particle they would often go on to find the times correctly. It was this vital first step that many candidates were unable to take. A few candidates made errors solving the quadratic equation once they had obtained it correctly.

Q9.

The candidates found this question more difficult than expected and there were many candidates who could not give correct responses to parts (c) and (d). While there were many good diagrams, some candidates did not label them clearly or did not include the 300 N force explicitly. The vast majority of candidates were able to gain all of the marks in part (b), but some did obtain the printed answer without sufficient working.

Part (c) was the most difficult part, with many candidates unsure which part of the train to consider. When candidates did consider the first carriage they often failed to include the 550 N force. The candidates did a little better on part (d), with those who considered the whole train generally being successful. Those who considered only the engine tended to omit the 1100 N force or not to use their answer from part (c).

Q10.

In part (a) many candidates lost marks for not including arrow heads or due to poor labelling of the forces. Several candidates did not show the two tensions as being equal on their diagrams. There were quite a few good attempts at part (b), with attempts at resolving vertically. Part (c) proved to be very challenging, although there were a number of good solutions. A number of candidates gained an M1 mark for the use of the friction law. The most common errors were to omit one force in their equation or to not include the 'ma' term in their equation of motion.

Q11.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for ignoring the m term or ignoring the minus sign. There were many good answers to part (b), although some used inventive algebra. The solution to part (c) was usually correct, although candidates often forgot to obtain ± 1, for the square root of v, when they found the square-root of the equation obtained in part (b).

In part (d) candidates who integrated v to find the distance travelled often forgot the + d term which was required to be shown to be equal to zero.

Those who used integration with limits commonly used $\int_{16}^{1} (4-0.1t)^2\,dt$ forgetting that the integral was with respect to t so that $\int_{0}^{30} (4-0.1t)^2\,dt$ was required. A considerable number of candidates, in part (d), tried to use equations for motion with constant acceleration.

Q12.

The candidates generally did well with parts (a) and (b), which were fairly standard, but had more difficulties with the later parts. The main reasons that candidates lost marks in part (a) were because they made errors with the signs in the equations or because they did not form an equation for each particle. Those with sign errors in part (a) often did not obtain the correct tension in part (b).

In part (c), many errors were due to the candidates using the wrong values in the wrong places. For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96 while in part (c)(ii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of in part (c)(ii), while the time of 2 seconds was sometimes used in part (c)(ii) instead of in part (c)(i).

Q13.

This question was also done well by many candidates. There were relatively few errors on the force diagrams. In part (b)(ii), some candidates made errors, the most common being to multiply the 30 by 0.5 or subtract the 30 from 49. Part (c) caused difficulties for some candidates. The most common errors here were to attempt to include the weight or to omit the friction force.

Q14.

The candidates generally did well with parts (a) and (b), which were fairly standard, but had more difficulties with the later parts. The main reasons that candidates lost marks in part (a) were because they made errors with the signs in the equations or because they did not form an equation for each particle. Those with sign errors in part (a) often did not obtain the correct tension in part (b).

In part (c), many errors were due to the candidates using the wrong values in the wrong places.

For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96, while in parts (c)(ii) and (c)(iii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of the later parts, while the time of 2 seconds was sometimes used in the later parts instead of in part (c)(i).

In part (c)(iii), the most common approaches involved the use of constant acceleration equations. While some candidates used the correct numerical values in their equations, many made errors with the signs. Another common approach was to find two times and add them together. With this method also, sign errors were often made by the candidates.

Q15.

While some candidates did well with this question, others made some fundamental errors. In part (a), some candidates found the magnitude of the acceleration and then tried to use this scalar quantity in the later parts of the question, ending up with a mix of scalar and vector quantities.

In part (b)(i), many candidates found the position vector, but did not go on to find the distance as requested. There were quite a number of correct responses to part (b)(ii), but quite often this was not followed by a correct approach to the final part of the question.

Q16.

Virtually all candidates answered part (a) correctly. The common problem in part (b) was in the integration of $400\cos\frac{\pi}{2}t$. Apart from this, answers to part (b) were good, and most correctly found $t = 2$ in part (c). The majority of candidates also found correctly that the speed of the particle was 3, with only a few finishing with the velocity of the particle being –3 or –3**i**.

Q17.

There were many good solutions to part (a). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach “lost” a negative sign in their working and ended up with a positive acceleration.

In part (b) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an acceleration of 9.8 instead of 0.98.

Q18.

The printed answer in part (a) seemed to help the candidates as there were very few incorrect answers to this part of the question. In part (b), while there were many good responses, some candidates used Speed = $\frac{\text{Distance}}{\text{Time}} = \frac{0.9}{0.6} = 1.5$ and then applied the equation $v = u + at$ to obtain an acceleration of 2.5. There were some very good answers to part (c), but some candidates gave answers such as "The accelerations are different." but did not explain why. In the better explanations, candidates made clear references to friction forces or air resistance.

Q19.

Parts (a) and (b), about the assumptions, were generally answered well. The most common incorrect answer was to say that the peg was fixed in part (a).

There were many good solutions to part (c). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach "lost" a negative sign in their working and ended up with a positive acceleration.

In part (d) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an acceleration of 9.8 instead of 0.98.

Q20.

Parts (a) and (b) were generally done well by the candidates. A few candidates subtracted the vectors in part (a) instead of adding them. Also some tried to find the magnitude of each vector in part (a) and then lost several marks because they had very confused working.

In part (c) there were many attempts which involved the division of vectors. For example working such as: $m = \frac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}}$ = 6**i** + 6**j**. Candidates should be advised to consider the components involved and not to attempt to divide vectors.

There were a number of errors that appeared in part (d). One was to include an initial velocity, which was equal to the resultant force on the particle. A few candidates used incorrect formulae. In the final part of the question a large number of candidates gave the vector 3**i** + 4**j** as their final answer and did not find the magnitude in order to get the distance as requested.

Q21.

Most of this question was answered well. In part (a), the only difficulty was found in differentiating $2e^{\frac{1}{2}t}$. In part (b), many found the velocity of the particle to be $(e^{\frac{3}{2}} - 8)\mathbf{i}$ [or –3.52**i**], but did not find the speed of +3.52 m s^{-1}. Others gave the speed as 3.5 m s^{-1}, which was penalised. Those candidates who were successful in part (a) usually completed parts (c) and (d) correctly.

Q22.

Most candidates started from $F = ma$ and answered part (a) correctly. Others were penalised for starting with $\frac{dv}{dt} = -\frac{0.08v^2}{0.05}$. There were many good answers to part (b), with very few candidates being unable to obtain $v = \frac{15}{5 + 24t}$ from $\frac{1}{v} = 1.6t + \frac{1}{3}$.

Q23.

Parts (a) and (c) were done well, with almost all of the candidates gaining full marks on these parts. Part (b) was a little more demanding. There were a number of correct solutions, but the two most common errors were to subtract 400 from 660 to obtain 260 N or to include the weight of the motorcycle and rider in some way.

Q24.

Part (a) of this question was done very well, but part (b) caused difficulties for some candidates. The most common errors in part (b) were to include the weight of the car and trailer or to produce equations in which the wrong combinations of forces were used.

Q25.

The vast majority of candidates gained full marks on this question. Two types of errors were seen. One was to assume that the 3000 N and 600 N forces were perpendicular and to calculate the magnitude based on this assumption. The second was to introduce the weight of the car in some way.

Q26.

There were lots of good responses to this question, although part (e) was sometimes incorrect.

Part (a) was found to be very straightforward and the majority of the candidates gained the marks, even if they did not go on to do other parts of the question.

Part (b) was also done well, with the vast majority of candidates stating clearly the two equations of motion that they went on to solve. A few did use a method with only one equation, but fortunately this method was not common. Also, a few used the weight of the block instead of the friction force.

Part (c) was also done well, provided the candidates had an appropriate equation to use from part (b).

It was interesting that some candidates who had found parts (a) to (c) difficult were able to use the acceleration given and produce correct answers to parts (d) and (e). There were many good solutions to part (d), but in part (e) candidates often continued to use the acceleration as 1.225 m s^{-2} or took the initial velocity of the particle to be zero. Some candidates made both of these errors. The success rate for part (e) was lower than that for part (d).

Q27.

The candidates found part (a) of this question relatively straightforward, but part (b) was found to be very demanding, and very few correct solutions were seen. Candidates lost marks in part (a) mainly due to errors in the resolving: for example using sin 30° instead of cos 30° in part (a)(ii). A few candidates lost one mark because they gave their final answer for the coefficient of friction as 0.31 rather than 0.313.

In part (b), very few candidates realised that they needed to express the normal reaction in terms of the tension. This error resulted in solutions that were flawed. Most of the candidates who did express the normal reaction in terms of the tension went on to gain full marks for this part of the question, although a few did make arithmetic or algebraic errors.

Q28.

Virtually all candidates answered this question well, with only a few forgetting to find the magnitude of the force in part (c).

Q29.

The answer printed in part (a) enabled virtually all candidates to create their solution. However, a considerable minority were not convincing. Part (b) was well answered, with most candidates appreciating that the differential equation needed to be solved by separating variables. Some found the resulting integral, $\int v^{-\frac{3}{2}}\,dv$, challenging, and the $+c$ term was often omitted. In part (c), rather than using $\frac{2}{\sqrt{v}} = \lambda t + \frac{2}{3}$ to obtain t when $v = 4$, many used the last result, $v = \frac{36}{(2 + 3\lambda t)^2}$ when $v = 4$, and forgot to eliminate the negative root when calculating the square root of $4 = \frac{36}{(2 + 3\lambda t)^2}$.

Q30.

While there were many good responses to this question, candidates did make some errors and some did not show enough working in part (a). In part (b), there were a number of incorrect approaches. These included:

- Simply calculating the weight.
- Calculating the magnitude of the resultant force using $F = ma$.
- Making a sign error by producing an equation such as $70g - T = 70 \times 0.64$.

Part (c) was usually done well, although a few candidates did not appear to know how to calculate the average speed.

Q31.

Generally part (a) was done well, whilst part (b) caused more difficulties. The fact that the acceleration was given did seem to help many candidates, but some did not justify the minus sign. Applying the constant acceleration equation to find the distance was also done well, although a few candidates took $u = 0$ and $v = 4$. There were a number of candidates who used two constant acceleration equations, finding the time taken to stop as an intermediate step. A few of these candidates did not go on to find the distance.

In part (b), many of the candidates were able to find the magnitude of the friction force. In a few cases the candidates did not show how the reaction force was obtained. Full marks were not awarded if the candidates simply stated $R = 1.84$ without any justification. When calculating the acceleration there were many sign errors in candidates' equations and some candidates also included a"4" which appeared as if it was a force. While many candidates realised that the puck would remain at rest, only a relatively small number of them were able to give a convincing explanation. Many of the explanations contained vague statements and lacked precision and clarity.

Q32.

Part (a) was done very well and some candidates had clearly been helped by the presence of the printed answer. There were several examples of candidates who obtained the printed answer in part (b) showing enough supporting working. A value of 4000 was often introduced without any working to show that it had been obtained from an application of Newton's Second Law.

Several candidates found the tension in the rope first and then found the value of P. Some candidates were confused about when to include the tension in the rope. The responses to part (c) were variable, but those who had taken a systematic approach to part (b) usually did well. The answers for part (d) were very varied, with some very good explanations and other responses which lacked coherence.

Q33.

This question was generally answered well. The only common error was in part (b)(ii), where a number of candidates found **F** to be 36**i** + 6**j**, but did not find its magnitude.

Q34.

The candidates made good attempts at this question. Some found it difficult to get going in part (a). Most of the candidates who had difficulties had tried to consider particle A instead of particle B. Generally the candidates made good attempts at parts (b) and (c). In part (d), the main problems were for candidates who found forming an equation of motion difficult. Often, these candidates had also found part (a) difficult. Most candidates were able to make an attempt at part (e) using their answers to parts (c) and (d).

Q35.

This question was done very well by the vast majority of the candidates. Almost all gained full marks on part (a), but a few did make arithmetic errors. Again, in part (b), almost all the candidates found the acceleration correctly, although a few did give the answer $\frac{4}{3}$. Part (c) did cause a little more difficulty, although there were many correct responses. The two most common errors were either to assume that the tension was equal to the weight or to make a sign error and obtain an equation such as
$400g - T = 400a$.